

- Czelusta, G., Mielczarek, J. (2025). **From $SU(2)$ holonomies to holographic duality via tensor networks.** *Physical Review D*, 111(6), 066012.
- Czelusta, G., Verma, D. R., Wanjalkar, G. (2024). **Grover's algorithm on two-way quantum computer.** *arXiv preprint arXiv:2406.09450.*
- Czelusta, G., Mielczarek, J. (2023). **Quantum circuits for the Ising spin networks.** *Physical Review D*, 108(8), 086027..
- Miroszewski, A., Mielczarek, J., Szczepanek, F., Czelusta, G., Grabowski, B., Saux, B. L., Nalepa, J. (2023). **Cloud Detection in Multispectral Satellite Images Using Support Vector Machines With Quantum Kernels.** *arXiv preprint arXiv:2307.07281.*
- Miroszewski, A., Mielczarek, J., Szczepanek, F., Czelusta, G., Grabowski, B., Saux, B. L., Nalepa, J. (2023). **Optimizing Kernel-Target Alignment for cloud detection in multispectral satellite images.** *arXiv preprint arXiv:2306.14515.*
- Miroszewski, A., Mielczarek, J., Czelusta, G., Szczepanek, F., Grabowski, B., Le Saux, B., Nalepa, J. (2023). **Detecting Clouds in Multispectral Satellite Images Using Quantum-Kernel Support Vector Machines.** *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing.*
- Czelusta, G. (2022). **Quantum computations in loop quantum gravity.** *In Particle Physics Summer Student Alumni Conference 2022 (Kraków, 9–10 July 2022) (p. 45).*
- Czelusta, G., Mielczarek, J. (2022). **Quantum variational solving of the Wheeler-DeWitt equation.** *Physical Review D*, 105(12), 126005.
- Czelusta, G., Mielczarek, J. (2021). **Secure quantum communication through a wormhole,** *arXiv:2103.14996*
- Czelusta, G., Mielczarek, J. (2021). **Quantum simulations of a qubit of space.** *Physical Review D*, 103(4), 046001.
- Ambjørn, J., Czelusta, G., Gizbert-Studnicki, J., Görlich, A., Jurkiewicz, J., Németh, D. (2020). **The higher-order phase transition in toroidal CDT.** *Journal of High Energy Physics*, 2020(2002.01051), 1-18.